

SANJAY KUMAR S

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EDUCATION

SRM University, KTR, Chennai

Chennai, India

B.Tech in Computer Science and Engineering (AI & ML)

2023 – 2027

- Current CGPA: 9.47

EXPERIENCE

Undergraduate Researcher – SRM UROP Program

Chennai, India

SRM Institute of Science and Technology

November 2025 – Present

- Developed reinforcement learning policies for autonomous drone navigation using Unreal Engine 5 and Cosys AirSim.
- Designed reward functions, state representations, and evaluation metrics for simulated navigation tasks.
- Optimized simulation runtime performance and environment interaction pipelines.
- Conducted experimentation on high-fidelity simulation environments for autonomous systems research.

TECHNICAL SKILLS

Programming: Python, Dart (Flutter), C, C++, Java, TypeScript

Frameworks & Tools: React 18, Vite, Linux, MySQL, Git, Unreal Engine 5, Cosys AirSim, Isaac Sim

Systems: Linux System Administration, Performance Optimization, Kernel Fundamentals, UNIX, RBAC Systems

AI / ML: Machine Learning, Deep Learning, Reinforcement Learning, Computer Vision

CERTIFICATIONS

SAP Certified – Generative AI Developer

[View Credential](#)

2026

Course on Python – LIVEWIRE

[Show Credential](#)

Sept 2023

Course on Java – LIVEWIRE

[Show Credential](#)

Aug 2024

Linux Programming – SRMIST

[Show Credential](#)

Aug 2024

PROJECTS

GigShield (CampTech)

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Full-Stack • AI/ML • Cloud Integration

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- **Tech Stack:** Flutter (Riverpod) / FastAPI (SQLAlchemy, JWT, bcrypt) / SQLite (PostgreSQL-ready) / Open-Meteo, OpenWeather AQI, OpenRouteService APIs / Blockchain-ready adapter.
- **Deployment:** Android APK build available for testing; backend architecture designed for scalable PostgreSQL-ready deployment workflows. [Download Android APK](#).
- AI-powered parametric insurance platform for gig workers. Engineered dynamic risk assessment by integrating live weather, traffic, and AQI APIs. Built a full-stack application with automated claims processing and fraud detection.
- [View Repository](#)

LLLMao - Local Large Language Model assistant orchestrator

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Backend • DevOps • Local AI

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- **Tech Stack:** React (TypeScript, Vite, Tailwind, Zustand) / FastAPI (SQLAlchemy, Pydantic, ChromaDB, SQLite) / Tauri (Rust) / Ollama (local inference, GPU acceleration) / PyPDF, python-docx, Watchdog.
- **Deployment:** Desktop-first local application; exploring Ubuntu Snap packaging for simplified Linux distribution.
- Developing a privacy-first local AI workstation integrating Ollama, FastAPI, ChromaDB, and a Tauri-based desktop UI with local RAG workflows and Linux-focused distribution exploration.
- [View Repository](#)

Aero-Controllers

Reinforcement Learning • Robotics • Simulation

- **Tech Stack:** Next.js (App Router) / React / Three.js + @react-three/fiber & drei / GitHub Pages.
- **Research Core:** Unreal Engine 5.5 / Cosys-AirSim / PPO (Stable-Baselines3).
- **Deployment:** Interactive documentation live on GitHub Pages.
- **Documentation repository for a validated research output:** an RL-based stabilization framework for autonomous drones in GPS-denied environments. Built a high-fidelity simulation in Unreal Engine 5 with AirSim and trained a PPO model to correct sensor drift.
- [View Repository](#) | [Live Demo](#)

Cattle-Go

Computer Vision • Deep Learning • AgriTech

- **Tech Stack:** Flutter (cross-platform) / FastAPI (Python) / TensorFlow/Keras (MobileNetV2) / Firebase Authentication & Firestore / Hugging Face Spaces / Ollama (RAG chatbot) / Multilingual UI (English, Tamil, Hindi).
- **Deployment:** [API Status \(service running\)](#) – [Download Android APK](#).
- AI livestock breed classification system using computer vision and deep learning techniques for Indian cattle breeds. Includes real-time detection, RAG chatbot integration, and multilingual support.
- [View Repository](#)

Padayappa-Paarvai

Frontend • Developer Tool • Visualization

- **Tech Stack:** React 19 / Vite / TypeScript / ES Modules / GitHub Pages / gh-pages.
- **Deployment:** [Live Demo \(GitHub Pages\)](#).
- A web-based repository visualizer that enables developers to explore large codebases confidently without backend dependencies.
- Features interactive tree visualization, advanced search and filtering workflows, automatic file and folder statistics, and fully client-side operation.
- [View Repository](#)

Achievements		
SRM SDG Hackathon – 3 rd Place		Chennai
Organized by SRMIST		Nov 2024
NLH AI x Legal Hackathon – 2 nd Place		Chennai
Madras Institute of Technology (Anna University)		Aug 2025

Languages	
◦ English, Tamil, Hindi (Read & Write), Telugu (Elementary Proficiency)	

Interests	
◦ Systems Engineering, Linux Infrastructure, Autonomous Systems	
◦ Kernel Development, Performance Optimization, Local AI Systems	